WORKED SOLUTION

STEP I 1987, Question 5

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q5

The key idea

The substitution is built so that both factors under the square root become perfect squares: $x-\alpha$ turns into a multiple of $\sin^2\theta$ and $\beta-x$ into the same multiple of $\cos^2\theta$. The square root then cancels exactly against $\mathrm{d}x/\mathrm{d}\theta$, and the first integral collapses to a constant integrand. The second integral keeps one extra factor $1/x$, which becomes a standard $\tan\theta$ integral once we divide top and bottom by $\cos^2\theta$.

Solution

The first integral, with $\alpha < \beta$

Put $x = \alpha\cos^2\theta + \beta\sin^2\theta$. Using $\cos^2\theta + \sin^2\theta = 1$,

$$x-\alpha = \alpha\cos^2\theta + \beta\sin^2\theta - \alpha(\cos^2\theta+\sin^2\theta) = (\beta-\alpha)\sin^2\theta,$$
$$\beta-x = \beta(\cos^2\theta+\sin^2\theta) - \alpha\cos^2\theta - \beta\sin^2\theta = (\beta-\alpha)\cos^2\theta,$$

and

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\alpha\cos\theta\sin\theta + 2\beta\sin\theta\cos\theta = 2(\beta-\alpha)\sin\theta\cos\theta.$$

For the limits: $\theta = 0$ gives $x = \alpha$ and $\theta = \frac{\pi}{2}$ gives $x = \beta$. Since $\mathrm{d}x/\mathrm{d}\theta > 0$ on $(0, \frac{\pi}{2})$, as θ runs over $[0, \frac{\pi}{2}]$, x runs once, increasingly, over $[\alpha, \beta]$.

On this range $\sin\theta \geq 0$ and $\cos\theta \geq 0$, and $\beta - \alpha > 0$, so

$$\sqrt{(x-\alpha)(\beta-x)} = \sqrt{(\beta-\alpha)^2\sin^2\theta\cos^2\theta} = (\beta-\alpha)\sin\theta\cos\theta.$$

Therefore

$$\int_\alpha^\beta \frac{\mathrm{d}x}{\sqrt{(x-\alpha)(\beta-x)}} = \int_0^{\pi/2} \frac{2(\beta-\alpha)\sin\theta\cos\theta}{(\beta-\alpha)\sin\theta\cos\theta}\,\mathrm{d}\theta = \int_0^{\pi/2} 2\,\mathrm{d}\theta = \pi,$$

as required. (The integrand blows up at both ends, so strictly this is an improper integral. The substitution shows it converges, because the transformed integrand is the bounded constant 2.)

Ask yourself Where exactly did we use $\alpha < \beta$?

Only in taking the square root: $\sqrt{(\beta-\alpha)^2}$ equals $\beta-\alpha$ precisely because $\beta-\alpha>0$. In general it is $|\beta-\alpha|$, and that is what drives the next part.

The first integral, with $\alpha > \beta$

Now the integral runs from the larger number α down to the smaller number β. On the interval between them, $x-\alpha<0$ and $\beta-x<0$, so the product is still positive and the integrand still makes sense.

The substitution goes through word for word, except that now $\sqrt{(\beta-\alpha)^2}=\alpha-\beta$, so

$$\frac{2(\beta-\alpha)\sin\theta\cos\theta}{(\alpha-\beta)\sin\theta\cos\theta}=-2,$$

and the integral is

$$\int_0^{\pi/2}(-2)\,\mathrm{d}\theta=-\pi.$$

Here is a quicker way to see it. The expression $(x-\alpha)(\beta-x)$ is unchanged when we swap α and β, so the integrand is the same function as in the first case with the roles of the two letters exchanged. Reversing the limits changes the sign, so

$$\int_\alpha^\beta = -\int_\beta^\alpha = -\pi,$$

using the first result with β as the smaller endpoint.

The second integral, with $0 < \alpha < \beta$

Exactly the same substitution, with the same cancellation, leaves only the factor $1/x$:

$$\int_\alpha^\beta \frac{\mathrm{d}x}{x\sqrt{(x-\alpha)(\beta-x)}}=\int_0^{\pi/2}\frac{2\,\mathrm{d}\theta}{\alpha\cos^2\theta+\beta\sin^2\theta}.$$

Because $\alpha,\beta>0$, the denominator is positive throughout $[0,\frac{\pi}{2}]$, so this is a proper integral. To evaluate it, divide numerator and denominator by $\cos^2\theta$ (valid for $0\le\theta<\frac{\pi}{2}$):

$$\int_0^{\pi/2}\frac{2\sec^2\theta\,\mathrm{d}\theta}{\alpha+\beta\tan^2\theta}.$$

Now put $t=\tan\theta$, so $\mathrm{d}t=\sec^2\theta\,\mathrm{d}\theta$, and t runs from 0 to ∞ as θ runs from 0 up to $\frac{\pi}{2}$:

$$\begin{aligned}\int_0^\infty\frac{2\,\mathrm{d}t}{\alpha+\beta t^2}&=\frac{2}{\beta}\int_0^\infty\frac{\mathrm{d}t}{t^2+\alpha/\beta}=\frac{2}{\beta}\sqrt{\frac{\beta}{\alpha}}\left[\arctan\left(t\sqrt{\frac{\beta}{\alpha}}\right)\right]_0^\infty\\&=\frac{2}{\sqrt{\alpha\beta}}\cdot\frac{\pi}{2}=\frac{\pi}{\sqrt{\alpha\beta}},\end{aligned}$$

as required. We needed $\alpha>0$ twice: so that $1/x$ has no singularity inside $[\alpha,\beta]$, and so that $\sqrt{\beta/\alpha}$ is real.

Ask yourself The upper limit became ∞. Is that a problem?

No. The θ-integral is perfectly ordinary; the infinite limit only appears because $\tan\theta \to \infty$ as $\theta \to \frac{\pi}{2}$. The t-integral converges, since arctan tends to $\frac{\pi}{2}$, and it equals the θ-integral as a limit of the integrals over $[0, \Theta]$ with $\Theta \to \frac{\pi}{2}$.

Checks

- Scaling: replacing x, α, β by kx, $k\alpha$, $k\beta$ leaves the first integral unchanged and divides the second by k. Both answers, π and $\pi/\sqrt{\alpha\beta}$, behave this way.
- Let $\alpha \to \beta$. Then $1/x \approx 1/\beta$ across the whole (tiny) interval, so the second integral should be about π/β, and indeed $\pi/\sqrt{\alpha\beta} \to \pi/\beta$.
- Numerically, with $\alpha = 2$, $\beta = 7$, quadrature gives 0.83963 for the second integral, matching $\pi/\sqrt{14} = 0.83963$. Random choices of $\alpha < \beta$ all gave π for the first integral, and swapping the endpoints gave $-\pi$.

Common slips

- Answering π again when $\alpha > \beta$. The integrand is positive, but the limits run backwards, so the answer is $-\pi$.
- Writing $\sqrt{(\beta-\alpha)^2\sin^2\theta\cos^2\theta} = (\beta-\alpha)\sin\theta\cos\theta$ without checking the signs of the three factors.

Beyond STEP

The substitution has a neat geometric meaning. Since $x = \frac{\alpha+\beta}{2} - \frac{\beta-\alpha}{2}\cos 2\theta$, it parametrises the semicircle on diameter $[\alpha, \beta]$, and $\sqrt{(x-\alpha)(\beta-x)}$ is the height of that semicircle above x. The second integral is a first taste of evaluating integrals by contour integration: in complex analysis, both results drop out of the residue theorem in a couple of lines. Needham's *Visual Complex Analysis* explains this beautifully, and the careful treatment of improper integrals is in Abbott's *Understanding Analysis*.

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